

Microprocessors And Microcontrollers

Assignment 4- Week 4

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

QUESTION 1:

Correct Answer: b

Detailed Solution:

QUESTION 2:

Correct Answer: c

Detailed Solution:

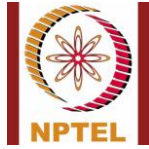
The **Interrupt Vector Table (IVT) of the 8085 microprocessor** is a fixed memory location range where the addresses of the interrupt service routines (ISR) are stored.

Interrupt Vector Table (IVT) Range in 8085

- The **8085 processor** has **vectored interrupts** that jump to predefined memory locations.
- Each vectored interrupt has a **fixed 8-bit address**, meaning it must be within the first **256 bytes** of memory.
- These addresses range from **0000H to 00FFH**.

Breakdown of Interrupt Vector Addresses:

Interrupt	Vector Address
TRAP	0024H
RST 7.5	003CH
RST 6.5	0034H
RST 5.5	002CH



Interrupt	Vector Address
INTR (if vectored externally) User-defined	
RST 0	0000H
RST 1	0008H
RST 2	0010H
RST 3	0018H
RST 4	0020H
RST 5	0028H
RST 6	0030H
RST 7	0038H

QUESTION 3:

Correct Answer: b

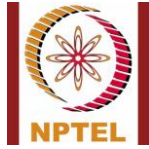
Detailed Solution:

When RST 7 instruction encountered corresponding ISR starting at memory location $7 \times 8 = (56)_{10} = (38)_{16}$ is executed.

QUESTION 4:

Correct Answer: c

Detailed Solution:



QUESTION 5:

INTR must remain active for how many T-states?

- A) 16.5 T-states
- B) 17.5 T-states
- C) 18.5 T-states
- D) 19.5 T-states

Correct Answer: B

Detailed Solution:

QUESTION 6:

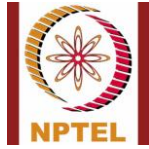
Multiply 03H and B2H (two 8-bit numbers) stored in memory locations 2200H and 2201H by repetitive addition and store the result in memory locations 2300H and 2301H. The output in 2300H and 2301H is

- a) 04H and 16H
- b) 16H and 02H
- c) 03H and 16H
- d) 16H and 03H

Correct Answer: b

Detailed Solution: (2200H) = 03H ; (2201H) = B2H

Result = B2H + B2H + B2H = 216H ; (2300H) = 16H ; (2301H) = 02H



QUESTION 7:

Correct Answer: a

Detailed Solution:

QUESTION 8:

Correct Answer: b

Detailed Solution:

QUESTION 9:

Correct Answer: c

Detailed Solution:

QUESTION 10:

Correct Answer: b

Detailed Solution:

QUESTION 11:

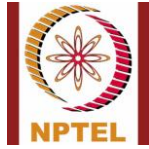
Correct Answer: b

Detailed Solution:

QUESTION 12:

Correct Answer: c

Detailed Solution:



QUESTION 13:

Correct Answer: a

Detailed Solution:

Bit 4 of the accumulator in the SIM instruction allows explicitly resetting the RST 7.5 memory

QUESTION 14:

Correct Answer: b

Detailed Solution:

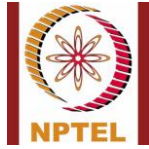
QUESTION 15:

How many 8-bit characters can be transmitted per second over a 9600 baud serial communication link using asynchronous mode of transmission with one start bit, eight data bits, two stop bits, and one parity bit?

- a) 600
- b) 800
- c) 1000
- d) 1200

Correct Answer: b

Detailed Solution: "9600 baud" means that the serial port is capable of transferring a maximum of 9600 bits per second.



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Total Data To send = 1 bit(start) + 8 bits (char size) + 1 bit (Parity) + 2 bits (Stop) = 12 bits.

Number of 8-bit characters that can be transmitted per second = $9600/12 = 800$.

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